Sample-Guided, Rotation-Robust Few-Shot Anomaly Detection and Localization for Semiconductor Wafer Dies with Feedback-Driven Adaptation

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1 Problem statement

Fully supervised inspectors need many labelled examples of every defect type, yet a new product or process step yields **few labelled defects**, and **defect types absent from training** fall outside a closed-set classifier's outputs. Learned features are also tied to the **orientation, position and scale** seen in training, so a rotated or shifted die can raise a false alarm or hide a defect, the motivation for registration-based few-shot methods [5, 6]. An image-level label does not say **where** the defect is, and pixel masks are costly to draw. Large public sets such as WM-811K are **discrete wafer maps** (one pass/fail cell per die), not die-surface images [7], so alone they cannot support pixel-level localization.

2 Proposed solution

An engineer supplies $K = 1$–8 defect-free sample dies of the same design. Each new die is aligned to them and compared patch by patch in a pretrained feature space, giving an anomaly map and a decision: **normal, known defect type or unknown anomaly**. Engineer-verified decisions update the reference memory; an update is kept only if held-out results do not degrade.

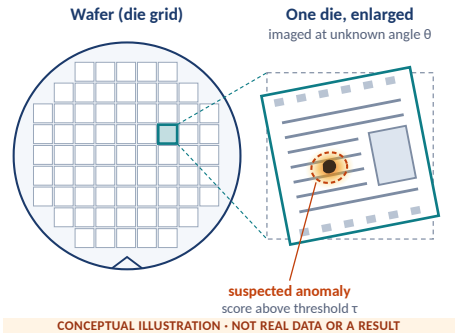


Fig. 2. Schematic only. The system marks *where* a die looks abnormal, not *why*; root cause needs failure analysis.

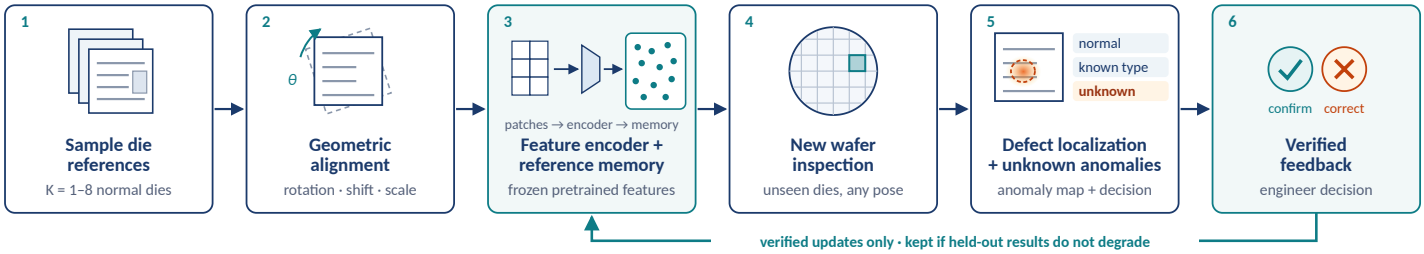


Fig. 1. Proposed inspection workflow (planned, not yet built). New dies share the alignment and encoder used for the references; feedback changes only the teal memory.

3 Research objectives

- **O1** **Few-shot detection** from $K \le 8$ normal references, no retraining.
- **O2** **Rotation robustness** to die rotation, shift, scale, illumination.
- **O3** **Localization:** pixel-level anomaly maps that match expert masks.
- **O4** **Unknown anomalies** kept apart from normal and known defect types.
- **O5** **Verified feedback** that improves later inspections, without forgetting.

4 Proposed method

- **Geometric alignment**: rotation, shift and scale vs. the references (feature matching + RANSAC or Fourier–Mellin); rotated references as fallback.
- **Pretrained features**: frozen patch features, DINOv2 ViT or CNN [1, 4].
- **Memory & scoring**: patch bank; nearest-neighbour distance per patch [4].
- **Localization**: upsampled score map; τ set on held-out normals.
- **Optional classification**: defect prototypes; far from all → **unknown**.

5 Datasets · raw surface images and discrete wafer maps are different data, used for different questions

Dataset	What it contains	Role in this study	Limitations
Carinthia-S [8] core · semiconductor	SEM images of **unpatterned** wafers (one layer); 4,591 images, 6 defect classes, expert binary masks.	Core test of pixel localization and known classes; held-out classes act as unknowns.	No repeated die layout or documented defect-free set (references from clean areas); licence unconfirmed.
WM-811K [7] wafer-level context	**Wafer maps, not images**: 811,457 per-die pass/fail maps; 172,950 labelled (9 classes).	Wafer-level few-shot and held-out-pattern tests; choosing which dies to inspect.	No die-surface appearance and **no pixel-level defect masks**.
MVTec AD [9] method check	5,354 industrial images, 15 categories, pixel masks.	Reproduce published baselines to check the implementation.	Not semiconductor data; CC BY-NC-SA 4.0.

Gap: no public, mask-annotated image set of *patterned* dies was found; rotation tests use rotated copies plus a small synthetic die set, reported as synthetic.

6 Evaluation plan

- **Detection**: image AUROC, AUPR, false-alarm rate at 95 % recall.
- **Localization**: pixel AUROC, AUPRO, IoU/Dice at τ.
- **Few-shot**: $K = 1, 2, 4, 8$; 5 random reference draws, mean ± s.d.
- **Rotation**: 0–360° (15° steps), shift, ±10 % scale; ± alignment.
- **Unknown**: leave-one-class-out; unknown-detection AUROC.
- **Feedback**: simulated stream incl. wrong labels; gain, forgetting.
- **Baselines**: PatchCore [4], RegAD [5], GraphCore [6], AnomalyDINO [1], UniVAD [10], IC segmentation [2]; supervised U-Net as upper bound.

7 Expected contribution (hypotheses, to be validated)

H1 At $K \le 8$, alignment plus rotated references reduces the accuracy loss under die rotation versus the same memory without alignment.

H2 Verified feedback written to the memory improves later localization without retraining or forgetting.

VS. PRIOR WORK Memory scoring [1, 4], registration [5], rotation-invariant features [6] and feedback memory correction [3] (MVTec AD, simulated feedback) exist separately. New here is their **joint, reproducible test on semiconductor SEM data** under rotation, open-set and feedback conditions; negative results count too.

8 References · [1]–[3] selected recent work (2025–2026) · [4]–[10] baselines and datasets

[1] S. Damm, M. Laszkiewicz, J. Lederer, A. Fischer, "AnomalyDINO: Boosting patch-based few-shot anomaly detection with DINOv2," *WACV*, 2025. CVF open access

[2] B. Zhao, Q. Shi, S. Lyu, Y. Lu, "Unsupervised integrated-circuit defect segmentation via image-intrinsic normality," arXiv:2509.09375, 2025 (preprint). doi:10.48550/arXiv.2509.09375

[3] A. Abbas, S. Abbas, K. Adhikari, "Training-free human-in-the-loop anomaly detection via memory bank correction," arXiv:2608.17775, 2026 (preprint). doi:10.48550/arXiv.2608.17775

[4] K. Roth, L. Pemula, J. Zepeda *et al.*, "Towards total recall in industrial anomaly detection" (PatchCore), *CVPR*, 2022. doi:10.1109/CVPR52688.2022.01392

[5] C. Huang, H. Guan, A. Jiang *et al.*, "Registration based few-shot anomaly detection" (RegAD), *ECCV*, 2022. doi:10.1007/978-3-031-20053-3_18

[6] G. Xie, J. Wang, J. Liu *et al.*, "Pushing the limits of fewshot anomaly detection in industry vision: GraphCore," *ICLR*, 2023. arXiv:2301.12082

[7] M.-J. Wu, J.-S. R. Jang, J.-L. Chen, "Wafer map failure pattern recognition and similarity ranking for large-scale data sets" (WM-811K), *IEEE Trans. Semicond. Manuf.*, 28(1), 2015. doi:10.1109/TSM.2014.2364237

[8] KAI GmbH & Univ. of Sarajevo, "Carinthia-S" (SEM images + masks), v1, Zenodo, 2025. zenodo.org/records/16895427

[9] P. Bergmann, M. Fauser, D. Sattlegger, C. Steger, "MVTec AD — A comprehensive real-world dataset for unsupervised anomaly detection," *CVPR*, 2019. doi:10.1109/CVPR.2019.00982

[10] Z. Gu, B. Zhu, G. Zhu *et al.*, "UniVAD: A training-free unified model for few-shot visual anomaly detection," *CVPR*, 2025. CVF open access